

BAHAA MOHAMMAD

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Iowa City, IA | U.S. Citizen | Languages: English, Arabic



EDUCATION

University of Iowa

Master of Science in Electrical and Computer Engineering, Accelerated BS/MS

Bachelor of Science in Engineering in Biomedical Engineering

Expected May 2027 • GPA 3.34

May 2026 • GPA 3.15

WORK EXPERIENCE

Software Engineer

June 2026 – Present

Voss Lab (HBC), University of Iowa | Iowa City, IA

- Built a **Python/R** pipeline (**JATOS** → auto **GGIR** → **REDCap** QC → debrief reports) for **295** subjects, replacing spreadsheet triage in cohort review.
- Automated **JATOS** pulls, **Globus** transfers, and **HPC** job submission so weekly pipelines sync and process **110 GB+** of multi-source study data end-to-end.
- Shipped QC/debrief scripts that turn large study dumps into repeatable, versioned lab review workflows.

Systems Engineer

May 2025 – Present

Human Technology (Santos) | Iowa City, IA

- Designed an **Oracle S3**-backed **C#** release/updater for **5** simulation apps with **GitHub Actions** CI/CD (merge → runner rebuild → version/size JSON → S3 publish); shared common assets cut combined install from **~7.5 GB** to a **2.2 GB** cap (**~70%**).
- Packaged delivery with **Docker**; migrated UI rendering to **C++** Diligent Engine and wired **C#/.NET** native interop so release builds exercise the full render path; cleared cross-platform defects and confirmed regression coverage before shipping each **GitHub Actions** release to **Oracle S3**.

Software Engineer

April 2024 – March 2026

Virtual Soldier Research (Santos DHM), University of Iowa | Iowa City, IA

- Owned day-to-day development on a **140+**-project **C++/C#/.NET/Fortran** Santos suite used daily by the research software team, shipping multi-language features into production research builds.
- Built and maintained **GitLab CI/CD** across the multi-language monorepo so merge builds stayed green, and kept Santos kinematics-to-graphics pipelines aligned so **C++/C#/.NET** visualization matched live simulation state.

RESEARCH EXPERIENCE

Research Software Engineer

March 2026 – May 2026

I-STRIKE (Santos contract), University of Iowa | Iowa City, IA

- Embedded a **MuJoCo** physics update loop inside Santos (**215** DOFs) via legacy **C#**/**C++** interop; fixed zero-mass bodies, frame mismatches, and anchors so contact simulation loads and runs.

Undergraduate Researcher

August 2025 – December 2025

University of Iowa Medical Imaging Lab | Iowa City, IA

- Built a multithreaded **Python/BART** MRI reconstruction pipeline across **4** GPUs that cut reconstruction time by about **5×**, with automated ingest and validation for clean experiment inputs.

PROJECTS

AlamirAuto Web Platform

TypeScript, React, Node.js, MongoDB

- Deployed a full-stack dealership marketplace with **REST** APIs, catalog filters, role-based admin inventory, auth, and automated backups, isolating stock updates from the customer-facing storefront.

SlamNav: GPS-Denied Indoor Mapping

Kotlin Multiplatform, C++, Swift, Objective-C, RTAB-Map

- Shipped a cross-platform Android/iOS navigation app (labeled **3D** maps, meshes, and routes without GPS) with a shared **C++ RTAB-Map** core, **Kotlin Multiplatform** UI, **JNI/Objective-C** bridges, and native **ARCore/ARKit** render pipelines.

Arteria: Urban Traffic Chokehold App

C#, .NET 8, Avalonia, Mapsui, OSM, SQLite

- Implemented **OSM** chokehold analysis in **C#** (Brandes betweenness, Hopcroft–Tarjan bridges, detour importance) plus historical traffic prediction and **SQLite** study stores; ran **GitHub Actions** `dotnet test` CI on every push.

PUBLICATIONS

- Abdel-Malek et al. & **Mohammad, B.** (2025). Santos DHM / Army ACFT. 9th Int. DHM Symp.
- Ali, M. S., **Mohammad, B.**, et al. (2025). Self-supervised spiral VAE, upper-airway MRI. ISMRM Workshop.